N-COIN: Stateless Consensus Currency

A New Value Exchange Standard Based on Seven-Party Unanimous Consensus

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Abstract

This paper proposes N-COIN, a stateless digital currency that maintains no historical ledger. N-COIN combines device-rooted UID keys, a seven-party unanimous protocol (SUP), MuSig2 signature aggregation, VRF-based mediator selection, and a trust-score mechanism to achieve lightweight and secure value transfer. By eliminating historical data and relying solely on the latest state, N-COIN avoids the storage bloat inherent in blockchain systems and enables operation even on smartphones.

1. Introduction

Modern digital financial systems are centralized, with user balances stored in institutional databases. Such systems inherently carry risks of censorship, freezing, and arbitrary policy changes.

Blockchain technology partially mitigates these issues, but PoW and PoS require large historical datasets and high operational costs. Majority-based consensus also tends to concentrate power among wealthy participants.

N-COIN introduces a stateless currency model. Instead of maintaining a global ledger, N-COIN updates only the latest state through a seven-party unanimous protocol (SUP). Combined with device-rooted UID keys and trust-score incentives, N-COIN enables secure and lightweight value exchange.

2. UID and State Structure

2.1 UID (Universal Identity)

Each N-COIN user is identified by a UID derived from a private key generated inside the device's Secure Enclave. The private key never leaves the device; only the public key is shared.

UID properties:

- Device-unique and difficult to clone
- Private key extraction is infeasible
- Device migration is supported via 2-of-2 signatures from the old device
- Independent of centralized identity providers

2.2 State Structure

N-COIN maintains only a single "latest state" per user, consisting of:

- UID (public key)
- Balance
- Trust score
- Transaction index (monotonically increasing counter)
- State hash
- Seven-party aggregated signature (MuSig2)

The state hash is computed as:

StateHash = Hash(UID | Balance | TrustScore | Index)

No historical data is stored. Only the latest state defines the user's value.

3. Seven-Party Unanimous Protocol (SUP)

3.1 Participants

Each transaction involves seven parties:

- A (sender)
- B (receiver)
- Five mediators selected via VRF

Mediator selection:

Mediator = VRF(UID | Index | RandomSeed)

3.2 Unanimity Requirement

A transaction is valid only if all seven participants sign it. MuSig2 aggregates the signatures into a single compact signature.

3.3 MuSig2 Verification

$g^s = R * (X_{agg} ^ c)$

- s: aggregated signature
- R: sum of all nonces
- X_agg: aggregated public key
- c: challenge hash

4. Transaction Protocol

4.0 Participation Requirement (Critical Rule)

A node may participate in a transaction only if its local state hash matches the latest state hash observed on the P2P network. Nodes with inconsistent states cannot issue transactions nor sign others' transactions until synchronization is restored.

4.1 Transaction Initiation

1. A issues a payment request.
2. Five mediators are selected via VRF.
3. A seven-party "transaction lock" is created.
4. While locked, A cannot participate in other transactions.

4.2 State Verification

A, B, and C1–C5 exchange their latest states and verify:

- Signature validity
- Monotonicity of Index
- Balance consistency

Any mismatch aborts the transaction.

4.3 State Update Proposal

A's update:

A_new_balance = A_old_balance - (Amount + Fee)

Index_new = Index_old + 1

TrustScore = TrustScore + 1

B's update:

B_new_balance = B_old_balance + Amount

Index_new = Index_old + 1

TrustScore = TrustScore + 1

Mediator update:

Mediator_new_balance = Mediator_old_balance + Fee

Index_new = Index_old + 1

TrustScore = TrustScore + 1

4.4 Mediator Verification

Mediators verify:

- Balance deltas
- Index monotonicity
- Fee correctness
- State hash consistency

If valid, they sign.

4.5 Finalization

- All seven signatures are collected
- MuSig2 aggregates them
- The new state is broadcast
- Old states are discarded after verification

5. Double-Spend Prevention

5.1 Transaction Lock

The lock process:

1. A lock is created when A initiates a transaction.
2. A cannot join other transactions while locked.
3. Mediators and B verify and sign the state.
4. Once all signatures are collected, the state is updated.
5. The lock is released.
6. If timeout occurs, the lock is automatically released.

5.2 Lock Properties

- Represents current occupancy
- Prevents double spending without historical data
- Robust against network delays

6. Trust Score Model

Trust scores evolve as:

If TransactionSuccess : TrustScore = TrustScore + 1

If Timeout : TrustScore = TrustScore + 0

If FraudDetected : TrustScore = TrustScore - 100

Nodes with low trust scores are excluded from mediator selection. Honest behavior yields the highest long-term reward.

7. Related Work

Bitcoin maintains a full historical ledger, causing unbounded storage growth. N-COIN differs fundamentally by storing only the latest state. Ethereum uses an account model but still requires historical data. Algorand employs VRF-based committee selection, which N-COIN adapts to a unanimous seven-party model. MuSig2 provides efficient signature aggregation, reducing communication overhead. Stateless blockchain research (e.g., Verkle trees) aims to reduce state size, but N-COIN eliminates historical state entirely.

8. Security Model

N-COIN assumes at most one malicious mediator. Since all seven signatures are required, a single dishonest participant cannot finalize a fraudulent transaction. Nodes must match the network's latest state hash before signing, preventing forks and inconsistencies. Double spending is prevented by the lock mechanism and the monotonic Index.

N-COIN maintains no historical ledger, but each node temporarily stores up to seven generations of recent state hashes received during transactions. Only hash values are retained, and storage remains constant regardless of network size.

In the extremely rare case where all seven participants collude maliciously, the retained hashes allow a statistical "hash-sum consistency check" to verify global network integrity without reintroducing historical storage. This preserves the stateless design while enabling network-wide validation when necessary.

9. Performance Analysis

N-COIN's state size is fixed and independent of network size, enabling operation on smartphones. MuSig2 reduces seven signatures to one, minimizing bandwidth. VRF computation is lightweight, allowing fast mediator selection. The absence of historical data eliminates storage growth and simplifies synchronization.

10. Discussion

The seven-party unanimous model differs from majority-based consensus by requiring full agreement. This avoids scenarios where honest minorities are overridden by dishonest majorities. Combined with trust scores, N-COIN creates an incentive structure where honesty yields the highest reward. Stateless currency represents a new direction for digital money, addressing blockchain's long-standing scalability issues.

The seven-party unanimous model draws conceptual inspiration from Akira Kurosawa's Seven Samurai. Just as seven unnamed samurai protected a village through personal integrity rather than institutional authority, each of the seven participants in an N-COIN transaction upholds network correctness through mutual verification. In a stateless system without historical records, the honesty of individual participants becomes the foundation of security.

11. Conclusion

N-COIN presents a stateless value exchange system based on unanimous consensus and trust-based incentives. By eliminating historical data and relying solely on the latest state, N-COIN offers a lightweight, secure, and decentralized alternative to both blockchains and centralized financial systems.

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Appendix A: Packet Structures

Lock

{ UID, Index, SignatureBundle }

StateUpdate

{ UID, NewState, SignatureBundle }

SignatureBundle

{ R, s, X_agg }

Appendix B: Formal Definitions

VRF

VRF(UID | Index | Seed) → Mediator

StateHash

Hash(UID | Balance | TrustScore | Index)

N-COIN Logo

The N-COIN logo is based on the lowercase letter “n”, combined with a modified vertical stroke of the dollar sign to abstractly resemble the Japanese character “七” (seven). The two vertical strokes symbolize twin blades—Guardians—that protect the immutability of the network. They also represent the seven approvers whose mutual trust forms the basis of the SUP model. At first glance the symbol appears as “n”, but upon deeper reading it reveals “7”, expressing the duality of anonymity and transparency that defines N-COIN